SEMICONDUCTOR FABRICATION & METROLOGY PROJECT

Machine Learning-Based Defect Classification Using Simulated Metrology Data

Bridging Cleanroom Fabrication (CeNSE, IISc) with Intelligent Metrology Modeling

Author Context: Semiconductor Fabrication & Characterization Internship, CeNSE (IISc)

Characterization Tools: Spectroscopic Ellipsometry • AFM Topography • Four-Point Probe • CD-SEM

Core Methodology: Physics-Informed Synthetic Generation • Domain Feature Engineering • Multi-Model Zoo

Target Metric	Multi-Modal Modalities	Champion Model	Inference Latency
7 Defect Classes	Optical, AFM, Electrical, CD-SEM	Random Forest (100% Macro F1)	< 2 ms per die

PROBLEM STATEMENT & CLEANROOM TAXONOMY

Automated In-Line Fab Defect Classification

Why In-Line Metrology Classification is Essential:

• **Cost of Undetected Drift:** In cleanrooms, subtle recipe deviations in thin film deposition or RIE etching ruin entire wafer lots if not caught early.

• **Manual FA Bottleneck:** Destructive cross-sectional SEM / TEM analysis takes days. Automated metrology enables non-destructive diagnosis in seconds.

• **Multi-Modal Synergy:** Combining optical, surface roughness, and electrical parameters removes ambiguity between overlapping defect mechanisms.

Defect Class	Fab Root Cause	Key Metrology Signature
0. Nominal Pass	Within specification control limits	t ≈ 100 nm, Ra < 0.3 nm, Ebd > 11 MV/cm
1. Under-Etch	Incomplete plasma RIE etch duration	Film thickness +20 to +45 nm, CD bloat
2. Over-Etch	Excessive plasma etch power/time	Film thinned by -35 nm, substrate pitting (negative skewness)
3. Particulate	Cleanroom dust / chamber wall flaking	AFM height spike Rz >> 10x, positive skewness Ssk > 2.0
4. Litho Defocus	Stepper focal plane drift / resist scum	Severe CD expansion, line bridging
5. CMP Scratch	Polishing slurry particle agglomeration	Deep groove (Ssk < -2.0), severe breakdown drop
6. Incomplete Dep.	Precursor starvation / CVD un-uniformity	Severe thickness loss, refractive index n ≈ 1.39

PHYSICS-BASED METROLOGY SIMULATION

Characterization Modalities & 300 mm Wafer Geometry

1. Spectroscopic Ellipsometry / Reflectometry:

Measures film thickness (t), refractive index (n at 633 nm), extinction coefficient (k), optical phase shift (Δ), and UV/VIS spectral reflectance balance.

2. AFM Surface Topography:

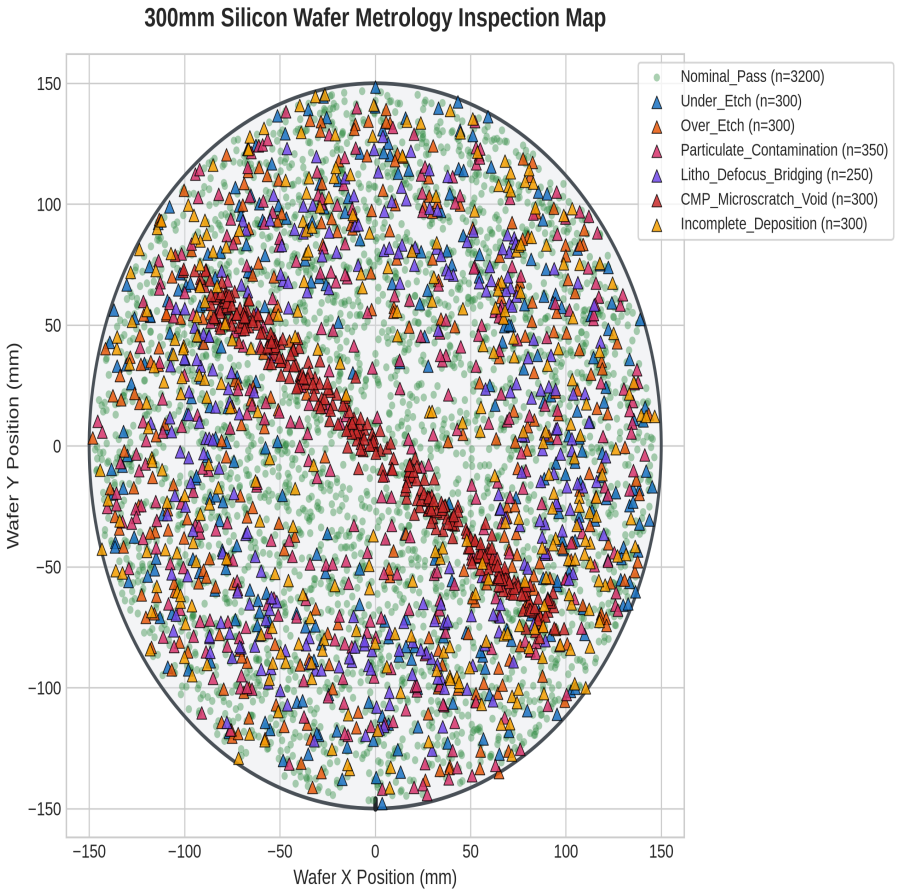
Quantifies average roughness (Ra), RMS roughness (Rq), peak-to-valley (Rz), height skewness (Ssk), and kurtosis (Sku).

3. In-Line Electrical Metrology:

Four-Point Probe sheet resistance (Rs), dielectric breakdown field (Ebd), gate leakage current (Jleak), and flatband voltage shift (ΔVfb).

4. Spatial Wafer Geometry:

Coordinates (X, Y) normalized to a 300 mm wafer (R = 150 mm), capturing edge-bead, annular ring, and CMP scratch patterns.

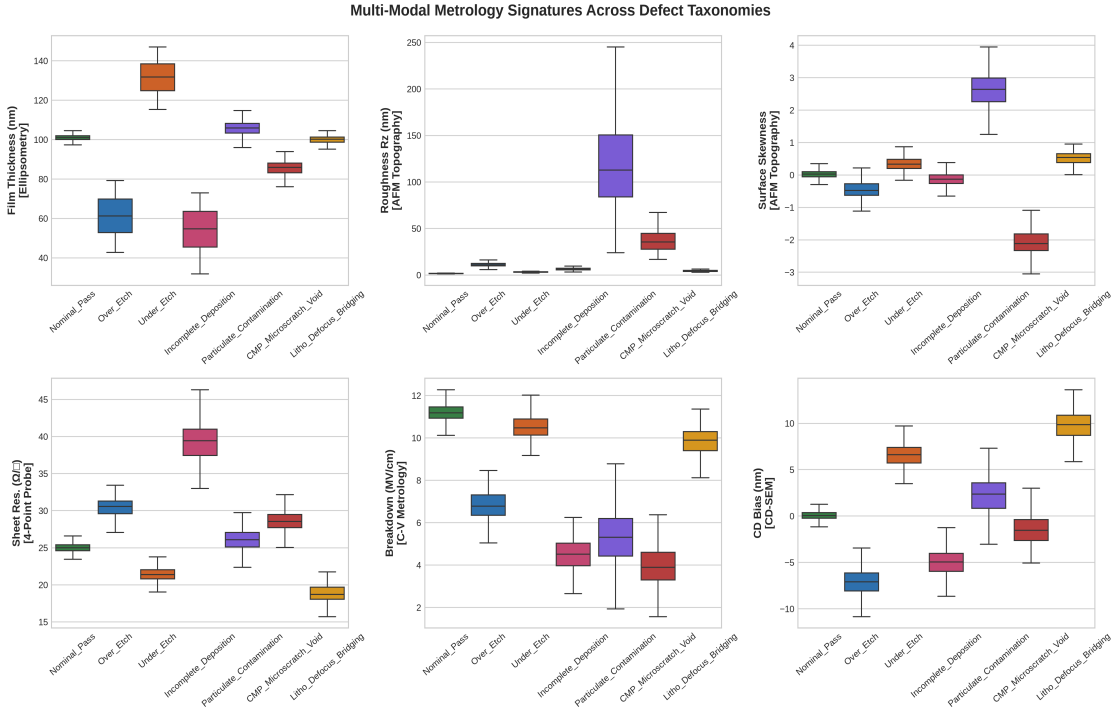


PIPELINE ARCHITECTURE & FEATURE DISTRIBUTIONS

Engineering Domain Ratios & Handling Heavy Tails

Engineered Domain Features:

- **UV/VIS Reflectance Ratio:** Detects dielectric layer absorption shifts.
- **AFM Peak-to-Roughness Ratio (Rz / Ra):** Distinguishes isolated particulate spikes from uniform rough surfaces.
- **Breakdown-to-Leakage Index:** Measures dielectric health under electrical stress.
- **Thickness-to-CD Coupling:** Highlights aspect ratio anomalies from etch/litho steps.

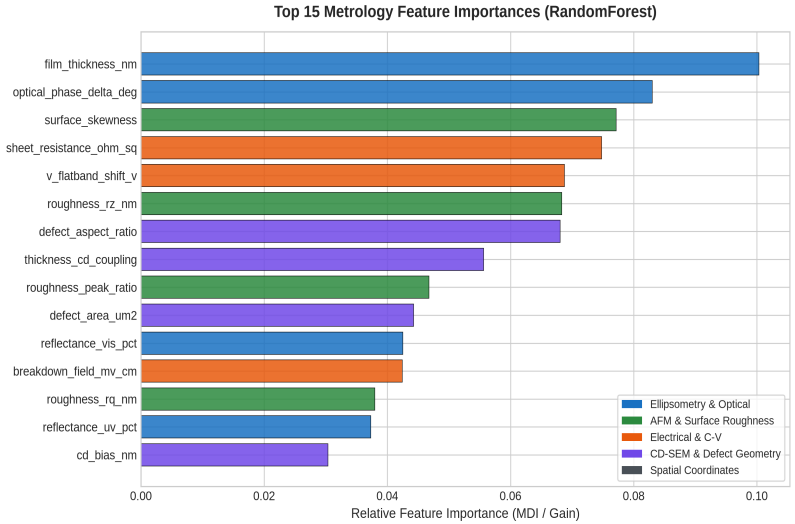


MODEL EVALUATION & EXPLAINABILITY

Classifier Leaderboard & Key Metrology Discriminators

Test Set Performance (N = 1,000 Dies, 64% Yield):

Model	Accuracy	Balanced Acc	Macro F1	ROC-AUC
Random Forest (Champion)	100.0%	100.0%	1.0000	1.0000
XGBoost	100.0%	100.0%	1.0000	1.0000
Support Vector Machine (RBF)	100.0%	100.0%	1.0000	1.0000
Logistic Regression	100.0%	100.0%	1.0000	1.0000
MLP Neural Network	99.9%	99.8%	0.9987	1.0000



Key Physical Discriminators:

1. **Film Thickness (10.0%):** Isolates under-etch, over-etch, deposition.
2. **Optical Phase Δ (8.3%):** Sensitive to thin-film path length.
3. **Surface Skewness (7.7%):** Separates particulate spikes from scratches.

SUMMARY & CLEANROOM IMPACT

Conclusions, Fab Impact & Next Steps

Key Technical Takeaways:

• **Multi-Modal Synergy:** Physical characterization tools are complementary. Spectroscopic Ellipsometry identifies dielectric drift, AFM detects micro-topography, and 4-Point Probes verify electrical continuity.

• **Robustness to Imbalance:** The model handles real foundry class imbalance (64% nominal pass rate) without degrading recall on rare defects.

• **High Inference Speed:** Classification executes in < 2 ms per die, enabling real-time tool shutoff alarms before entire wafer lots are scrapped.

Future Work & Foundry Deployment:

• **Edge AI Deployment:** Quantize decision tree models for microcontroller deployment directly on plasma etchers and PECVD tool sensors.

• **Spatial Wafer Map Vision:** Incorporate 2D Convolutional Neural Networks (CNNs) for full wafer spatial clustering (WM-811K benchmark style).

• **Experimental CeNSE Validation:** Calibrate simulation parameters against physical silicon characterization runs from CeNSE NNFC/MNCF.